Code No: 157FB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
UTILIZATION OF ELECTRICAL ENERGY
(Common to CE, ME, ECE, CSE)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain in brief how heating is done in the following cases:
i) Resistance heating ii) Induction heating iii) Dielectric heating
b) At full load of 10 h.p, the temperature rise of a motor is 25^0C after one hour, and 40^0C after 2 hours. Find the final temperature rise on full load. Assume that the iron losses are 80% of full load copper losses. [9+6]

2.a) What are the advantages of electrically produced heat? What are the properties to be possessed by the element used in resistance oven?
b) A piece of an insulating material is to be heated by dielectric heating. The size of the piece is 10×10×3 cm. A frequency of 20 Megacycles/sec is used and the power absorbed is 400W. Calculate the voltage necessary for heating and the current that flows in the material. The material has a relative permittivity of 5 and a power factor of 0.05. [8+7]

3.a) Describe briefly the various types of arc welding processes used in industry.
b) A 40 kW, 3-phase, 400 volts resistance oven is to employ Ni-Cr strip of 0.3 mm thickness. The heating elements are star connected. If the wire temperature is to be 1127^0C and that of charge is to be 727^0C, estimate the suitable width and length of the wire required. Radiation efficiency = 0.6, Specific resistance of Ni-Cr = 1.03×10^{-6} ohm-m. Emissivity = 0.9. [8+7]

4.a) Explain the principle of electrolysis.
b) Write a short notes on electromagnetic stirs. [8+7]

5.a) What are the essentials of a good lighting system? What do you understand by glare?
b) A drawing hall 30 m × 15 m with a ceiling height of 5 m has to be provided with general illumination of 120 lux. Taking utilization of 0.5, depreciation of 1.4, determine the number of fluorescent lamps required, their spacing, mounting height and total wattage. Take luminous efficiency of fluorescent tube of 40 lumens/watt for 80 watt tubes. [8+7]

6. Give the construction and working of the following types of lamps:
a) Mercury Vapour lamp
b) Sodium Vapour lamp [8+7]

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7.a) For a trapezoidal speed-time curve of an electric train, derive expression for maximum speed and distance between stops.

b) A mail is to be run between two stations 5kms apart at an average speed of 50km/hr. If the maximum speed is to be limited to 70km/hr, acceleration to 2km/hr/sec, braking retardation to 4km/hr/sec and coasting retardation to 0.1km/hr/sec, determine the speed at the end of coasting, duration of coasting period and braking period. [8+7]

8.a) Compare single battery system with double battery parallel block system.

b) Explain the special requirements for train lighting. [8+7]

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